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# Towards Reactive Electrochemical Transport Using the Lattice Boltzmann Method. Part I: Electrostatics, Hydrodynamics, Species Transport, and Electrokinetic Benchmarking

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## Abstract

This work presents the first stage in the development of a lattice Boltzmann framework for reactive electrochemical transport. The present study focuses on the verification and validation of electrostatics, hydrodynamics, ionic species transport, and coupled electrokinetic phenomena using analytical benchmark solutions. A modular lattice Boltzmann framework is developed in Fortran, in which independent distribution functions are employed to solve the Poisson equation, Navier–Stokes equations, and Nernst–Planck equations within a unified computational framework. The numerical solutions are systematically compared with the corresponding analytical solutions, demonstrating excellent agreement for the electrostatic potential, electro-osmotic velocity, ionic concentration profiles, and the fully coupled electrokinetic system. The benchmark presented in this work establishes a robust and extensible numerical foundation for future developments toward reactive electrochemical transport and multiscale electrocatalytic simulations.

## Keywords

Lattice Boltzmann Method, Electrokinetics, Poisson–Boltzmann Equation, Nernst–Planck Equation, Reactive Mass Transfer, Benchmark

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**Abstract**

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## 1 Introduction

Electrochemical energy conversion and storage technologies have emerged as indispensable components of the transition toward sustainable energy systems. Processes such as the electrochemical carbon dioxide reduction reaction ($CO_2$RR), nitrogen reduction reaction ($N_2$RR), nitrate reduction reaction ($NO_3$RR), oxygen reduction reaction (ORR), oxygen evolution reaction (OER), hydrogen evolution reaction (HER), hydrogen oxidation reaction (HOR), and electrochemical biomass conversion have attracted considerable attention because of their potential to produce valuable chemicals and fuels while reducing greenhouse gas emissions and enabling carbon-neutral energy cycles. Despite their diverse chemistries, all of these electrocatalytic processes are governed by the same fundamental physical phenomena, namely electrostatics, ionic migration, molecular diffusion, hydrodynamic transport, heterogeneous reaction kinetics, and their mutual coupling across multiple length and time scales[1–3].

Among these reactions, $CO_2$RR represents one of the most challenging and extensively studied examples because it simultaneously addresses carbon utilization and renewable fuel production. However, its practical implementation remains hindered by low current densities and poor product selectivity[4–6]. These limitations originate not only from the intrinsic catalytic properties of the electrode but also

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from the restricted transport of reactants and products. In aqueous electrolytes, the low solubility and diffusivity of $CO_2$ lead to severe concentration gradients near the catalyst surface, while thick concentration boundary layers significantly reduce the reactant supply to the electrode[4–6]. Similar transport limitations are also encountered in other electrocatalytic systems involving poorly soluble reactants, most notably nitrogen reduction, where the extremely low solubility of $N_2$ in water poses an equally severe challenge for achieving industrially relevant reaction rates[7]. Consequently, understanding and controlling transport phenomena has become one of the central challenges in modern electrocatalysis.

Considerable efforts have therefore been devoted to improving mass transport through reactor design and operating conditions. Gas diffusion electrodes, flow-cell configurations, membrane electrode assemblies, and advanced electrolyte engineering have substantially increased attainable current densities by enhancing reactant delivery to the catalyst surface. Nevertheless, these improvements simultaneously modify the local electrochemical environment, altering ion distributions, electric fields, local pH, reactant concentrations, solvent organization, and hydrodynamic flow[4–6]. These quantities collectively define the local reaction environment (LRE), which ultimately governs the activity, selectivity, and stability of electrocatalytic processes.

The importance of the local reaction environment has been highlighted by numerous theoretical and experimental studies[4–7]. In $CO_2$RR, for example, electrolyte composition[8, 9], cation identity[8, 10], water activity[9], applied potential[11], and flow conditions[11, 12] have all been shown to exert a profound influence on catalytic performance. Several mechanisms have been proposed to explain these observations, including stabilization of reaction intermediates by electrolyte cations[8], modification of the local electric field[10], variations in proton availability[9], and changes in interfacial water structure[10]. Similar observations have been reported for many other electrocatalytic reactions[13–18], demonstrating that catalyst performance cannot be understood solely from the electronic structure of the catalyst itself. Instead, it emerges from the intricate coupling between heterogeneous surface reactions and the surrounding transport environment.

From a theoretical perspective, remarkable progress has been achieved in computational electrocatalysis over the past two decades. Density functional theory (DFT) has become the standard first-principles approach for investigating adsorption energetics, reaction mechanisms, elementary reaction pathways, and catalyst design at electrified solid–liquid interfaces [1, 19]. These atomistic simulations provide highly accurate information regarding the electronic structure and reaction energetics of catalytic surfaces. However, their computational cost restricts simulations to systems containing only a few hundred atoms and time scales of several tens of picoseconds. Consequently, atomistic models describe only the immediate vicinity of the catalyst surface and cannot capture the transport of reactants, ions, solvent molecules, and products over experimentally relevant spatial and temporal scales. Recent multiscale approaches combining DFT with microkinetic models and continuum transport descriptions have significantly improved the predictive capability of computational electrocatalysis[20]. Nevertheless, the accurate representation of the coupled electrostatic, hydrodynamic, and species transport phenomena remains one of the principal challenges in developing predictive electrochemical cell models.

At the continuum level, electrochemical transport is governed by a strongly coupled set of nonlinear partial differential equations describing electrostatics, fluid flow, and mass transport[21]. The simultaneous solution of the Poisson equation, Navier–Stokes equations, and Nernst–Planck equations rapidly becomes computationally demanding for realistic reactor geometries and operating conditions. Traditional numerical approaches based on finite-difference[22], finite-volume[23], and finite-element methods[24] have achieved considerable success; however, the increasing complexity of electrochemical systems motivates the exploration of alternative numerical frameworks capable of efficiently treating coupled multiphysics transport problems.

Among the available numerical techniques, the lattice Boltzmann method (LBM) has emerged as a powerful alternative for solving complex transport phenomena. Unlike conventional continuum solvers, LBM is formulated from the Boltzmann kinetic equation and naturally accommodates the coupling of multiple physical processes within a unified computational framework. Its explicit formulation, straightforward implementation of complex boundary conditions, excellent parallel scalability, and ability to bridge microscopic and macroscopic descriptions make it particularly attractive for multiscale electrochemical simulations[25, 26].

The long-term objective of the present work is to establish a comprehensive lattice Boltzmann framework capable of simulating reactive electrochemical transport in electrocatalytic systems. Ultimately, the proposed framework will integrate electrostatics, hydrodynamics, ionic species transport, heterogeneous reaction kinetics, and atomistically derived reaction energetics into a unified multiscale methodology suitable for investigating realistic electrochemical reactors.

The present manuscript constitutes Part I of this research program. In this first contribution, we establish the numerical foundations of the framework by developing and systematically validating each physical component of the model independently before coupling them into a unified electrokinetic formulation. Analytical benchmark solutions are employed to verify the electrostatic potential, hydrodynamic flow, ionic species transport, and the fully coupled electrokinetic model. These benchmark studies provide the essential verification step required before introducing heterogeneous electrochemical reactions and reactive mass transfer in subsequent parts of this series.

## 2 Governing Equations

To describe the coupled multiphysics transport phenomena, we first introduce the governing equations employed throughout this work.

### 2.1 Electrostatics

The electrostatic potential is governed by the Poisson equation[27, 28],

$$\nabla^2\psi = -\frac{\rho_e}{\varepsilon}, \qquad \varepsilon = \varepsilon_0\varepsilon_r, \qquad \rho_e = F\sum_i z_i c_i. \tag{1}$$

where $\rho_e$, $\varepsilon$, $F$, $z_i$, and $c_i$ denote the charge density, dielectric permittivity of the medium, Faraday constant, valence of ionic species $i$, and concentration of ionic species $i$, respectively.

Substituting the Boltzmann distribution for the ionic concentrations into Eq. (1) yields the nonlinear Poisson–Boltzmann equation[27, 28],

$$\nabla^2\psi = -\frac{F}{\varepsilon}\sum_i z_i c_{i,\infty}\exp\left(-\frac{z_i F\psi}{RT}\right). \tag{2}$$

where $R$, $T$, and $c_{i,\infty}$ denote the universal gas constant, absolute temperature, and the bulk concentration of ionic species $i$, respectively. Equation (2) is valid for an arbitrary multi-component electrolyte.

For a symmetric $z : z$ electrolyte with equal bulk concentrations of cations and anions, Eq. (2) simplifies to[27, 28]

$$\nabla^2\psi = \frac{2zFc_\infty}{\varepsilon}\sinh\left(\frac{zF\psi}{RT}\right), \tag{3}$$

which is the form employed in the present benchmark study.

For sufficiently small electrostatic potentials, Debye and Hückel linearized the Poisson–Boltzmann equation by expanding the hyperbolic sine function as a first-order Taylor series. The zeroth-order term vanishes because the electrolyte is electrically neutral in the bulk, leaving only the linear term. This approximation results in the linearized Poisson–Boltzmann equation[27, 28],

$$\nabla^2\psi = \kappa^2\psi, \qquad \kappa^2 = \frac{2z^2F^2c_\infty}{\varepsilon RT}. \tag{4}$$

For a symmetric channel with identical wall potentials, the analytical solution of Eq. (4) is given by [27, 28]

$$\psi(y) = \zeta \, \frac{\cosh\left[\kappa\left(y - \frac{H}{2}\right)\right]}{\cosh\left(\kappa \frac{H}{2}\right)}. \tag{5}$$

where $\zeta$ is the zeta potential at the channel walls and $H$ is the channel width.

## 2.2 Hydrodynamics

To describe the fluid flow in the microchannel, the Navier–Stokes equation is employed[21, 27, 28],

$$\rho\left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u}\cdot\nabla\mathbf{u}\right) = -\nabla p + \mu\nabla^2\mathbf{u} + \rho_e\mathbf{E_x}, \tag{6}$$

where $\mathbf{u}$ is the fluid velocity vector, $E_x$ is the externally applied electric field in the $x$ direction, $p$ is the pressure, $\mu$ is the dynamic viscosity, and $\rho$ is the fluid density.

The continuity equation for an incompressible fluid is given by[21]

$$\nabla\cdot\mathbf{u} = 0, \qquad \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0. \tag{7}$$

where $u$ and $v$ are the velocity components in the $x$ and $y$ directions, respectively.

For a fully developed electro-osmotic flow, Eq. (6) simplifies to[21, 27, 28]

$$0 = -\frac{dp}{dx} + \mu\frac{d^2u}{dy^2} + \rho_e(y)E_x. \tag{8}$$

Using the Poisson equation to eliminate the charge density, Eq. (8) can be rewritten as [21, 27, 28]

$$\mu\frac{d^2u}{dy^2} = \frac{dp}{dx} + \varepsilon E_x\frac{d^2\psi}{dy^2}. \tag{9}$$

For a purely electro-osmotic flow, where no external pressure gradient is applied, the analytical solution of Eq. (9) subject to the no-slip boundary condition at the channel walls is given by[21, 27, 28]

$$u(y) = -\frac{\varepsilon\zeta E_x}{\mu}\left[1 - \frac{\cosh\left[\kappa\left(y - \frac{H}{2}\right)\right]}{\cosh\left(\kappa\frac{H}{2}\right)}\right]. \tag{10}$$

Equation (10) represents the analytical solution for electro-osmotic flow obtained under the Debye–Hückel approximation, in which the Poisson–Boltzmann equation is linearized. Throughout this work, Eqs. (5) and (10) are employed as analytical benchmark solutions to validate the numerical solutions of the Poisson equation, Navier–Stokes equation, and their coupled electrokinetic formulation.

However, the Boltzmann distribution for the ionic concentration is valid only under thermodynamic equilibrium. In many electrochemical systems, the ionic concentrations deviate from the equilibrium distribution due to convection, diffusion, electromigration, and heterogeneous electrochemical reactions. In such cases, the ionic concentrations must be determined by solving the full species transport equation.

## 2.3 Species Transport

The conservation of species is described by[21, 27, 28]

$$\frac{\partial c_i}{\partial t} + \nabla\cdot\mathbf{J}_i = R_i. \tag{11}$$

where $R_i$ is the reaction rate of species $i$, and $\mathbf{J}_i$ is the molar flux. According to the Nernst–Planck equation, the molar flux is given by[21, 27, 28]

$$\mathbf{J}_i = c_i\mathbf{u} - D_i\nabla c_i - \frac{D_i z_i F}{RT}c_i\nabla\psi. \tag{12}$$

Here, $D_i$ is the diffusion coefficient of species $i$. Substituting Eq. (12) into Eq. (11) yields the species transport equation[21, 27, 28],

$$\frac{\partial c_i}{\partial t} + \nabla \cdot (c_i \mathbf{u}) = D_i \nabla^2 c_i + \frac{D_i z_i F}{RT} \nabla \cdot (c_i \nabla \psi) + R_i. \tag{13}$$

For a fully developed, non-reactive transport problem with zero wall-normal flux, the analytical solution of Eq. (13) is[21, 27, 28]

$$c_i(y) = c_{i,\mathrm{mid}} \exp\left[ -\frac{z_i F}{RT} \left( \psi(y) - \psi_{\mathrm{mid}} \right) \right]. \tag{14}$$

where $c_{i,\mathrm{mid}}$ and $\psi_{\mathrm{mid}}$ denote the ionic concentration and electrostatic potential at the centerline of the microchannel, respectively.

It is worth noting that, under the assumptions of thermodynamic equilibrium together with the present geometry and boundary conditions, the analytical solution of Eq. (13) is equivalent to the classical Boltzmann distribution given in Eq. (14). However, once these assumptions are relaxed, for example in the presence of heterogeneous reactions, concentration gradients, pressure-driven flow, or non-equilibrium transport conditions, the ionic concentration generally deviates from the Boltzmann distribution. Consequently, the full species transport equation must be solved to accurately describe the coupled electrochemical transport phenomena.

## 3 Lattice Boltzmann Formulation

The governing equations presented in the previous section are solved using the lattice Boltzmann method (LBM), which is a mesoscopic numerical technique derived from the Boltzmann transport equation. Unlike conventional continuum-based numerical methods, the LBM solves for the evolution of particle distribution functions on a discrete lattice. The macroscopic governing equations are subsequently recovered through appropriate velocity moments of these distribution functions[26].

In the present work, three independent distribution functions are employed to describe the coupled electrokinetic transport phenomena. The electrostatic potential is solved using the distribution function $h_\alpha$, the hydrodynamic equations are solved using the distribution function $f_\alpha$, while the transport of ionic species is described by the distribution function $g_\alpha$. Although each physical process is solved independently, they remain fully coupled through the electrostatic body force, ionic concentrations, and fluid velocity.

All three transport equations are discretized on a two-dimensional nine-velocity (D2Q9) lattice as shown in Fig. 1.

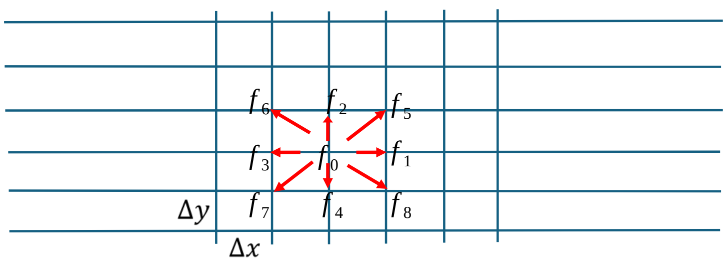


Figure 1: Schematic for two-dimensional nine velocity lattice.

### 3.1 Electrostatic Solver

The electrostatic potential is obtained by solving the Poisson equation using the lattice Boltzmann method following the method used by Chai and Shi[29]. The evolution equation for the electrostatic distribution function $h_i$ is written as

$$h_i(\mathbf{x} + \mathbf{c}_i \Delta t, t + \Delta t) - h_i(\mathbf{x}, t) = \Omega_i(\mathbf{x}, t) + \Omega_i'(\mathbf{x}, t), \tag{15}$$

where $\mathbf{c}_i$ denotes the discrete lattice velocity, while $\Omega_i$ and $\Omega_i'$ represent the collision and source operators, respectively. The Bhatnagar–Gross–Krook (BGK) approximation is adopted for the collision operator[29],

$$\Omega_i = -\frac{1}{\tau_h} \left( h_i - h_i^{eq} \right), \tag{16}$$

where $\tau_h$ is the relaxation time associated with the electrostatic solver and $h_i^{eq}$ is the corresponding equilibrium distribution function.

The equilibrium distribution function is defined as

$$h_i^{eq} = \begin{cases} (w_0 - 1)\psi, & i = 0, \\ w_i \psi, & i = 1, \dots, 8. \end{cases} \tag{17}$$

For the D2Q9 lattice, the standard weighting factors are given by

$$w_i = \begin{cases} \frac{4}{9}, & i = 0, \\ \frac{1}{9}, & i = 1, \dots, 4, \\ \frac{1}{36}, & i = 5, \dots, 8. \end{cases} \tag{18}$$

The source operator associated with the right-hand side of the Poisson equation is written as[29]

$$\Omega_i' = R_D \Delta t \, \overline{w}_i, \tag{19}$$

where $R_D$ represents the source term of the diffusion-form equation used to recover the electrostatic governing equation. The modified weighting factors employed for the source operator are

$$\overline{w}_i = \begin{cases} 0, & i = 0, \\ \frac{1}{8}, & i = 1, \dots, 8. \end{cases} \tag{20}$$

The use of $\overline{w}_0 = 0$ distributes the source contribution exclusively among the eight moving lattice directions, whereas the equilibrium distribution is constructed using the standard D2Q9 weights $w_i$.

Applying the Chapman–Enskog expansion[25, 30] shows that the above lattice Boltzmann formulation recovers the corresponding macroscopic diffusion-form equation, with the lattice diffusivity related to the relaxation time by

$$D = \frac{1}{3} c^2 \left( \frac{1}{2} - \tau_h \right) \Delta t, \qquad c = \frac{\Delta x}{\Delta t}, \tag{21}$$

where $c$ is the lattice speed, $\Delta x$ is the lattice spacing, and $\Delta t$ is the time step.

Finally, the electrostatic potential is recovered from the nonzero lattice populations according to[29]

$$\psi = \frac{1}{1 - w_0} \sum_{i=1}^{8} h_i. \tag{22}$$

### 3.2 Hydrodynamic Solver

The fluid flow is solved using the lattice Boltzmann distribution function $f_i$. The evolution of $f_i$ on the D2Q9 lattice is described by the discrete lattice Boltzmann equation[26, 30]

$$f_i\left(\mathbf{x}+\mathbf{e}_i\Delta t, t+\Delta t\right) - f_i(\mathbf{x},t) = -\frac{1}{\tau_f}\left(f_i - f_i^{eq}\right) + \Delta t\,\mathcal{F}_i, \tag{23}$$

where $\mathbf{e}_i$ is the discrete lattice velocity, $\tau_f$ is the hydrodynamic relaxation time, $f_i^{eq}$ is the equilibrium distribution function, and $\mathcal{F}_i$ is the discrete forcing term.

The second-order equilibrium distribution function is defined as[26, 30]

$$f_i^{eq} = w_i\rho\left[1+\frac{\mathbf{e}_i\cdot\mathbf{u}}{c_s^2}+\frac{\left(\mathbf{e}_i\cdot\mathbf{u}\right)^2}{2c_s^4}-\frac{\mathbf{u}\cdot\mathbf{u}}{2c_s^2}\right], \tag{24}$$

where $\rho$ and $\mathbf{u}$ denote the fluid density and velocity, respectively. For the D2Q9 lattice, the lattice speed of sound is given by[26, 30]

$$c_s^2 = \frac{1}{3}. \tag{25}$$

The corresponding D2Q9 weighting factors are[26, 30]

$$w_i = \begin{cases} \dfrac{4}{9}, & i = 0, \\ \dfrac{1}{9}, & i = 1,\ldots,4, \\ \dfrac{1}{36}, & i = 5,\ldots,8. \end{cases} \tag{26}$$

The electrostatic body force acting on the fluid is defined as[31]

$$\mathbf{F} = \rho_e\left(\mathbf{E}-\nabla\psi\right), \tag{27}$$

where $\mathbf{E}$ is the externally applied electric field and $-\nabla\psi$ is the electric field associated with the electrostatic potential inside the electrolyte. Therefore, the term $\mathbf{E}-\nabla\psi$ represents the total electric field acting on the local charge density.

The discrete forcing term employed in the lattice Boltzmann equation is written as

$$\mathcal{F}_i = \frac{\rho_e\left(\mathbf{E}-\nabla\psi\right)\cdot\left(\mathbf{e}_i-\mathbf{u}\right)}{\rho RT} f_i^{eq}. \tag{28}$$

This forcing formulation introduces the electrostatic force into the evolution of the hydrodynamic distribution function and consequently couples the electrostatic and fluid-flow solvers.

The hydrodynamic relaxation time is related to the kinematic viscosity $\nu$ through[26]

$$\tau_f = \frac{3\nu}{c\Delta x} + \frac{1}{2}, \tag{29}$$

where

$$c = \frac{\Delta x}{\Delta t} \tag{30}$$

is the lattice speed, $\Delta x$ is the lattice spacing, and $\Delta t$ is the time step. Equivalently, the viscosity relation can be written as[26]

$$\nu = c_s^2\left(\tau_f - \frac{1}{2}\right)\Delta t. \tag{31}$$

The macroscopic density is recovered from the zeroth moment of the distribution function[26],

$$\rho = \sum_i f_i, \tag{32}$$

while the macroscopic fluid velocity is obtained from the first moment. When the forcing contribution is included using a centered-force correction, the velocity is calculated as[26]

$$\rho \mathbf{u} = \sum_i f_i \mathbf{e}_i + \frac{\Delta t}{2}\mathbf{F}. \tag{33}$$

### 3.3 Species Transport Solver

The transport of each chemical species $a$ is solved using the lattice Boltzmann distribution function $g_i^{(a)}$ following the method introduced by He et al.,[31]. The evolution equation is written as

$$\begin{aligned} g_i^{(a)}\left(\mathbf{x} + \mathbf{e}_i \Delta t_{D_a}, t + \Delta t_{D_a}\right) - g_i^{(a)}(\mathbf{x}, t) = &-\frac{1}{\tau_{D_a}}\left[g_i^{(a)}(\mathbf{x}, t) - g_i^{(a),eq}(\mathbf{x}, t)\right] \\ &+ w_i \Delta t_{D_a}\left(1 - \frac{1}{2\tau_{D_a}}\right)\frac{e z_a D_a}{kT}\nabla \cdot \left(C_a \nabla \psi\right). \end{aligned} \tag{34}$$

where $\tau_{D_a}$ is the relaxation time associated with the transport of species $a$, $D_a$ is its diffusion coefficient, $z_a$ is its valence, $e$ is the elementary charge, $k$ is the Boltzmann constant, and $\delta_{t,D_a}$ is the time-step parameter used for the species-transport solver. The last term accounts for ionic migration induced by the electrostatic potential gradient.

The equilibrium distribution function is defined separately for the rest, axial, and diagonal lattice directions as[31]

$$g_i^{(a),eq} = \begin{cases} -\dfrac{2C_a u^2}{3c_{D_a}^2}, & i = 0, \\ \dfrac{C_a}{9}\left[\dfrac{3}{2} + \dfrac{3\mathbf{e}_i \cdot \mathbf{u}}{2c_{D_a}^2} + \dfrac{9\left(\mathbf{e}_i \cdot \mathbf{u}\right)^2}{2c_{D_a}^4} - \dfrac{3u^2}{2c_{D_a}^2}\right], & i = 1, \ldots, 4, \\ \dfrac{C_a}{36}\left[3 + \dfrac{6\mathbf{e}_i \cdot \mathbf{u}}{c_{D_a}^2} + \dfrac{9\left(\mathbf{e}_i \cdot \mathbf{u}\right)^2}{2c_{D_a}^4} - \dfrac{3u^2}{2c_{D_a}^2}\right], & i = 5, \ldots, 8. \end{cases} \tag{35}$$

Here, $C_a$ denotes the concentration of species $a$, $\mathbf{u}$ is the fluid velocity, and $c_{D_a}$ is the characteristic lattice speed associated with the species-transport solver.

The macroscopic concentration is recovered from the distribution functions with a correction associated with the electromigration source term[31],

$$C_a = \sum_{i=0}^{8} g_i^{(a)} + \frac{\Delta t_{D_a}}{2}\frac{e z_a D_a}{kT}\nabla \cdot \left(C_a \nabla \psi\right). \tag{36}$$

The relaxation time is related to the diffusion coefficient through[31]

$$D_a = c_{s,D_a}^2\left(\tau_{D_a} - \frac{1}{2}\right)\Delta t_{D_a}, \tag{37}$$

where $c_{s,D_a}$ is the corresponding lattice speed of sound for the species-transport formulation.

This formulation couples species transport to both the hydrodynamic and electrostatic solvers. The velocity field obtained from the hydrodynamic solver enters the equilibrium distribution function through the advective terms, while the electrostatic potential obtained from the Poisson solver contributes to ionic migration through the source term. Conversely, the calculated concentrations are used to update the local charge density appearing in the electrostatic governing equation.

# 4 Boundary Conditions

## 4.1 Electrostatic Boundary Conditions

A constant electrostatic potential was imposed at the upper and lower channel walls. The prescribed wall potential is denoted by $\psi_w$. In the lattice Boltzmann formulation, the Dirichlet boundary condition was implemented through an anti-bounce-back relation for the electrostatic distribution function $h_i$.

At the upper wall, located at $y = H$, the unknown incoming populations were reconstructed according to[26]

$$h_8(x, H) = (w_8 + w_6)\,\psi_w - h_6(x, H), \tag{38}$$
$$h_7(x, H) = (w_7 + w_5)\,\psi_w - h_5(x, H), \tag{39}$$
$$h_4(x, H) = (w_4 + w_2)\,\psi_w - h_2(x, H). \tag{40}$$

At the lower wall, located at $y = 0$, the corresponding relations are[26]

$$h_2(x, 0) = (w_4 + w_2)\,\psi_w - h_4(x, 0), \tag{41}$$
$$h_6(x, 0) = (w_8 + w_6)\,\psi_w - h_8(x, 0), \tag{42}$$
$$h_5(x, 0) = (w_5 + w_7)\,\psi_w - h_7(x, 0). \tag{43}$$

These relations enforce the prescribed wall potential while reconstructing the unknown distribution functions from their opposite lattice directions.

Periodic boundary conditions were applied in the streamwise $x$ direction. Accordingly, populations leaving one side of the computational domain re-entered from the opposite side. For the electrostatic distribution function, this periodic treatment can be written schematically as[26]

$$h_i(0, y) = h_i(L_x, y), \tag{44}$$

with the appropriate directional populations transferred between the left and right boundaries.

## 4.2 Hydrodynamic Boundary Conditions

The no-slip condition was imposed at the upper and lower channel walls using the bounce-back scheme for the hydrodynamic distribution function $f_i$. In this treatment, populations directed toward a solid wall are reflected into their opposite lattice directions, thereby enforcing zero fluid velocity at the wall.

At the lower wall, located at $y = 0$, the unknown outgoing populations were reconstructed as[26]

$$f_2(x, 0) = f_4(x, 0), \tag{45}$$
$$f_5(x, 0) = f_7(x, 0), \tag{46}$$
$$f_6(x, 0) = f_8(x, 0). \tag{47}$$

At the upper wall, located at $y = H$, the corresponding bounce-back relations are[26]

$$f_4(x, H) = f_2(x, H), \tag{48}$$
$$f_7(x, H) = f_5(x, H), \tag{49}$$
$$f_8(x, H) = f_6(x, H). \tag{50}$$

These relations reverse the wall-normal component of the particle populations and impose the no-slip boundary condition at both channel walls.

Periodic boundary conditions were applied in the streamwise $x$ direction. Populations leaving one side of the computational domain were transferred to the corresponding incoming directions at the opposite side. The directional relations used in the implementation are[26]

$$f_1(0,y) = f_1(L_x,y), \qquad f_5(0,y) = f_5(L_x,y), \qquad f_8(0,y) = f_8(L_x,y), \tag{51}$$

$$f_3(L_x,y) = f_3(0,y), \qquad f_6(L_x,y) = f_6(0,y), \qquad f_7(L_x,y) = f_7(0,y). \tag{52}$$

Equivalently, the macroscopic flow field satisfies periodicity in the streamwise direction,

$$\mathbf{u}(0,y) = \mathbf{u}(L_x,y). \tag{53}$$

## 4.3 Species Boundary Conditions

For the benchmark calculations, a zero wall-normal flux condition was imposed at the upper and lower channel walls. Under thermodynamic equilibrium, this condition is equivalent to prescribing the ionic concentration according to the Boltzmann distribution[32],

$$C_w = C_{\text{bulk}} \exp\left[-\frac{z_a e\left(\psi_w - \psi_{\text{mid}}\right)}{kT}\right], \tag{54}$$

where $C_w$ is the ionic concentration at the wall, $C_{\text{bulk}}$ is the bulk concentration, $z_a$ is the ionic valence of species $a$, $e$ is the elementary charge, $\psi_w$ is the electrostatic potential at the wall, and $\psi_{\text{mid}}$ is the electrostatic potential at the centerline of the channel.

The unknown lattice populations at the wall were reconstructed from the prescribed concentration. The zeroth lattice population was first obtained from[32]

$$C_0 = 3C_w - 3S_g - \frac{3}{2}S_{\text{source}}, \tag{55}$$

where $S_g$ denotes the summation of the known incoming distribution functions and $S_{\text{source}}$ is the electromigration source term evaluated at the wall.

The unknown outgoing populations at the lower wall were then reconstructed as[32]

$$g_2 = \frac{C_0}{6}, \qquad g_5 = \frac{C_0}{12}, \qquad g_6 = \frac{C_0}{12}, \tag{56}$$

whereas the unknown populations at the upper wall were obtained from[32]

$$g_4 = \frac{C_0}{6}, \qquad g_7 = \frac{C_0}{12}, \qquad g_8 = \frac{C_0}{12}. \tag{57}$$

This reconstruction guarantees that the prescribed equilibrium concentration is satisfied while simultaneously enforcing the zero normal-flux boundary condition at the solid walls.

Periodic boundary conditions were applied in the streamwise direction. Accordingly, the species distribution functions satisfy[26]

$$g_6(L_x,y) = g_6(0,y), \qquad g_5(0,y) = g_5(L_x,y), \tag{58}$$

$$g_3(L_x,y) = g_3(0,y), \qquad g_7(L_x,y) = g_7(0,y), \tag{59}$$

$$g_1(0,y) = g_1(L_x,y), \qquad g_8(0,y) = g_8(L_x,y). \tag{60}$$

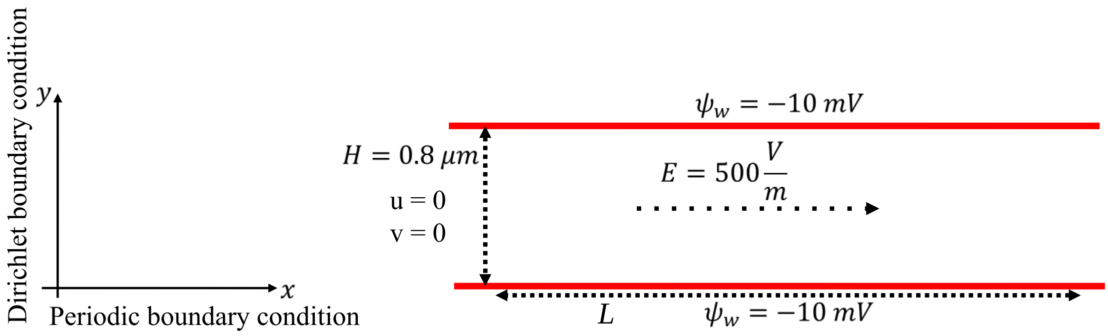


Figure 2: Geometrical domain and model system.

# 5 Numerical Method

The benchmark simulations were performed for a two-dimensional microchannel as shown in Fig. 2 with a physical height of

$$H = 0.80\ \mu\text{m}. \tag{61}$$

A symmetric electrolyte with a bulk concentration of

$$c_\infty = 0.10\ \text{mol}\,\text{L}^{-1} \tag{62}$$

was considered. Both channel walls were maintained at a constant electrostatic potential of

$$\zeta = -10\ \text{mV}. \tag{63}$$

The electrolyte was assumed to have the physical properties of liquid water, with a density of $\rho = 1000\ \text{kg}\,\text{m}^{-3}$, a kinematic viscosity of $\nu = 1.0 \times 10^{-6}\ \text{m}^2\,\text{s}^{-1}$, and a relative dielectric permittivity of $\varepsilon_r = 78.5$. An external electric field of $E_x = 5.0 \times 10^2\ \text{V}\,\text{m}^{-1}$ was applied in the streamwise direction, while no external electric field was applied in the wall-normal direction, such that $E_y = 0$.

The simulations were carried out at a temperature of $T = 273$ K. The physical parameters and fundamental constants employed in the calculations are summarized in Table 1.

Table 1: Physical parameters and fundamental constants used in the benchmark simulations.

| Parameter | Symbol | Value |
|---|---|---|
| Channel height | $H$ | $0.80 \times 10^{-6}$ m |
| Bulk electrolyte concentration | $c_\infty$ | $0.10\ \text{mol}\,\text{L}^{-1}$ |
| Wall potential | $\zeta$ | $-10.0 \times 10^{-3}$ V |
| Fluid density | $\rho$ | $1000\ \text{kg}\,\text{m}^{-3}$ |
| Kinematic viscosity | $\nu$ | $1.0 \times 10^{-6}\ \text{m}^2\,\text{s}^{-1}$ |
| External electric field in the $x$ direction | $E_x$ | $5.0 \times 10^2\ \text{V}\,\text{m}^{-1}$ |
| External electric field in the $y$ direction | $E_y$ | $0\ \text{V}\,\text{m}^{-1}$ |
| Vacuum permittivity | $\varepsilon_0$ | $8.8541878128 \times 10^{-12}\ \text{F}\,\text{m}^{-1}$ |
| Relative permittivity of water | $\varepsilon_r$ | 78.5 |
| Elementary charge | $e$ | $1.602176634 \times 10^{-19}$ C |
| Boltzmann constant | $k_\text{B}$ | $1.380649 \times 10^{-23}\ \text{J}\,\text{K}^{-1}$ |
| Temperature | $T$ | 273 K |
| Avogadro constant | $N_\text{A}$ | $6.02214076 \times 10^{23}\ \text{mol}^{-1}$ |

A uniform computational lattice consisting of $200 \times 400$ grid nodes was employed throughout the simulations. For the electrostatic solver, the relaxation time and lattice speed were set to $\tau_h = 1.0$ and

$c = 1.0$, respectively. The iterative solution was considered converged when the maximum relative error fell below $10^{-6}$.

For the hydrodynamic solver, a relaxation time of $\tau_f = 0.65$ was employed. The convergence criterion was defined as a maximum relative error of $3 \times 10^{-6}$.

For the species transport solver, the relaxation time was set to $\tau_D = 1.0$, and the iterative solution was terminated when the maximum relative error reached $10^{-15}$.

# 6 Validation

## 6.1 Electrostatic Benchmark

The numerical solution of the electrostatic potential was compared with the analytical solution given by Eq. (5). Excellent agreement was obtained.

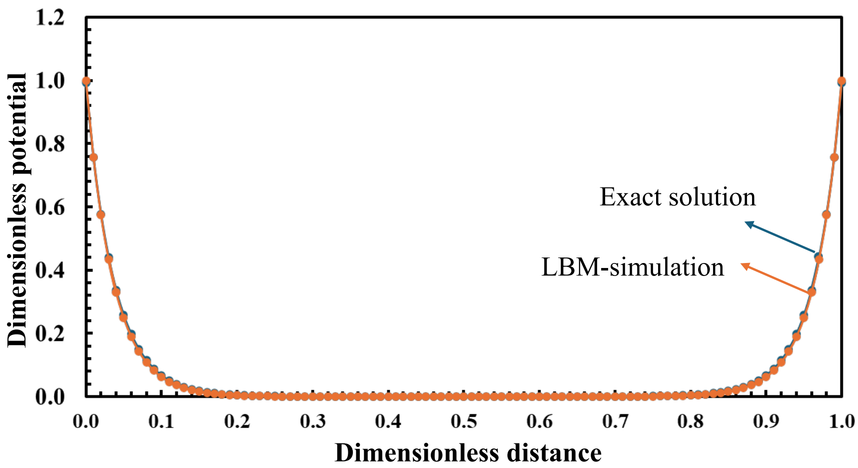


Figure 3: Comparison between the analytical and lattice Boltzmann solutions for the electrostatic potential.

As shown in Fig. 3, the numerical solution is almost indistinguishable from the analytical solution.

## 6.2 Hydrodynamic Benchmark

The numerical solution of the electro-osmotic velocity profile was compared with the analytical solution given by Eq. (10). Excellent agreement was obtained over the entire computational domain.

As shown in Fig. 4, the numerical solution is in excellent agreement with the analytical solution. The relative error is negligible throughout the channel, demonstrating that the proposed lattice Boltzmann formulation accurately reproduces the electro-osmotic flow field.

## 6.3 Species Transport Benchmark

The numerical solution of the ionic concentration profile was compared with the analytical solution given by Eq. (14). The benchmark was performed for both ionic species under the prescribed electrostatic potential and zero wall-normal flux boundary condition.

As shown in Fig. 5, the lattice Boltzmann solution accurately reproduces the analytical concentration profile for both ionic species. The excellent agreement confirms the correctness of the implemented Nernst–Planck solver and its coupling with the electrostatic field.

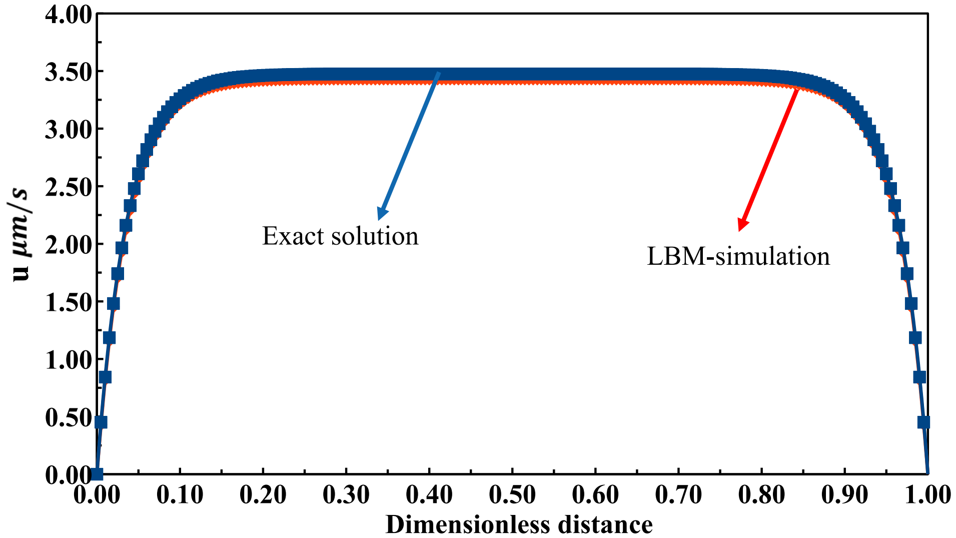


Figure 4: Comparison between the analytical and lattice Boltzmann solutions for the electro-osmotic velocity profile.

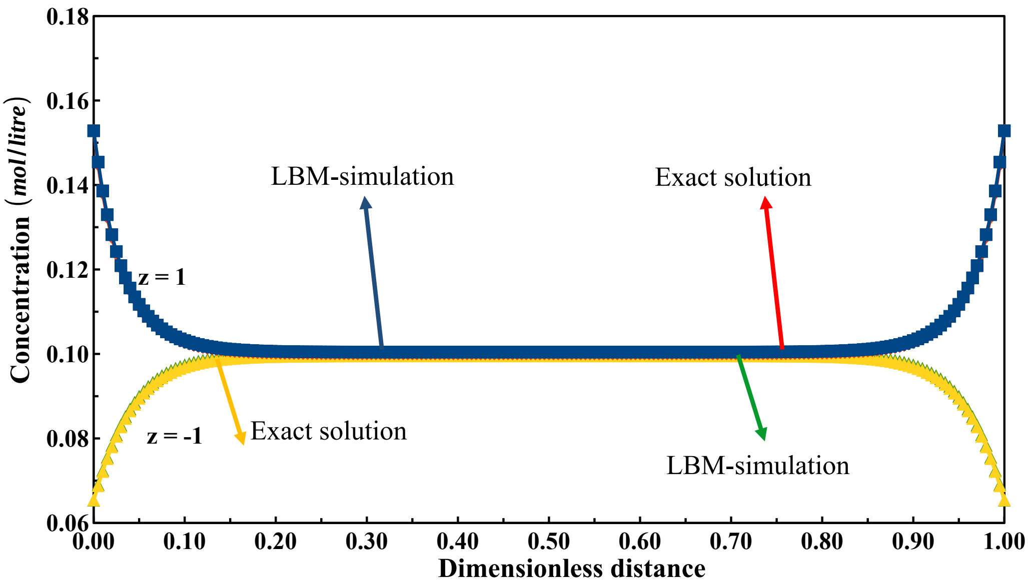


Figure 5: Comparison between the analytical and lattice Boltzmann solutions for the ionic concentration profile.

# 7 Results

To further verify the implementation of the proposed lattice Boltzmann framework, the electrostatic potential, electro-osmotic velocity, and ionic concentration fields are presented over the entire computational domain. In contrast to the one-dimensional benchmark comparisons presented in the previous section, these contour plots demonstrate that the prescribed boundary conditions are correctly imposed throughout the domain and that the coupled electrokinetic solver remains stable and physically consistent.

The electrostatic potential distribution is shown in Fig. 6. The corresponding electro-osmotic velocity field together with the streamlines is presented in Fig. 7, while the concentration distributions of the ionic species are shown in Fig. 8.

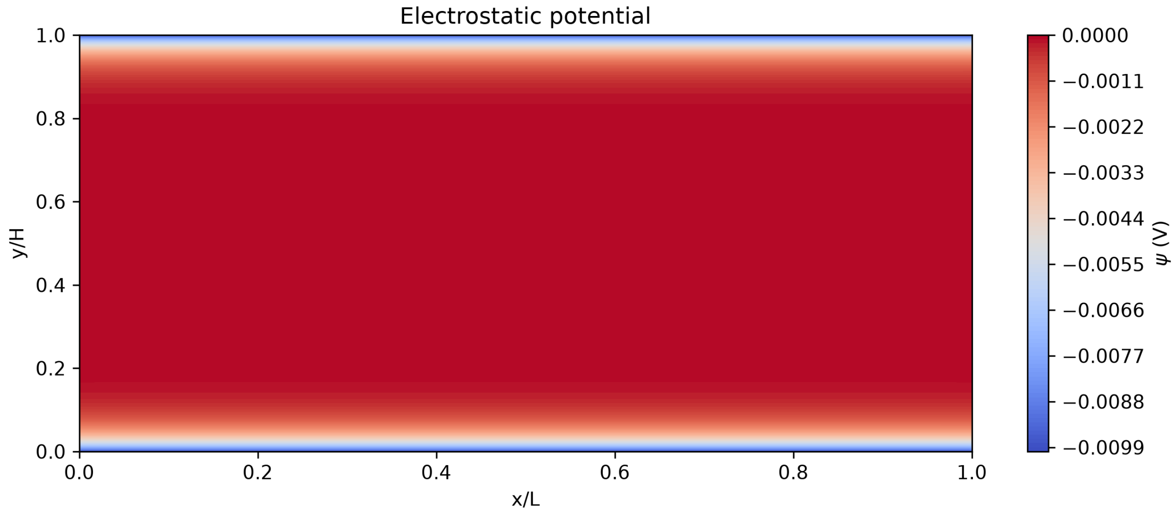


Figure 6: Contour distribution of the electrostatic potential within the computational domain.

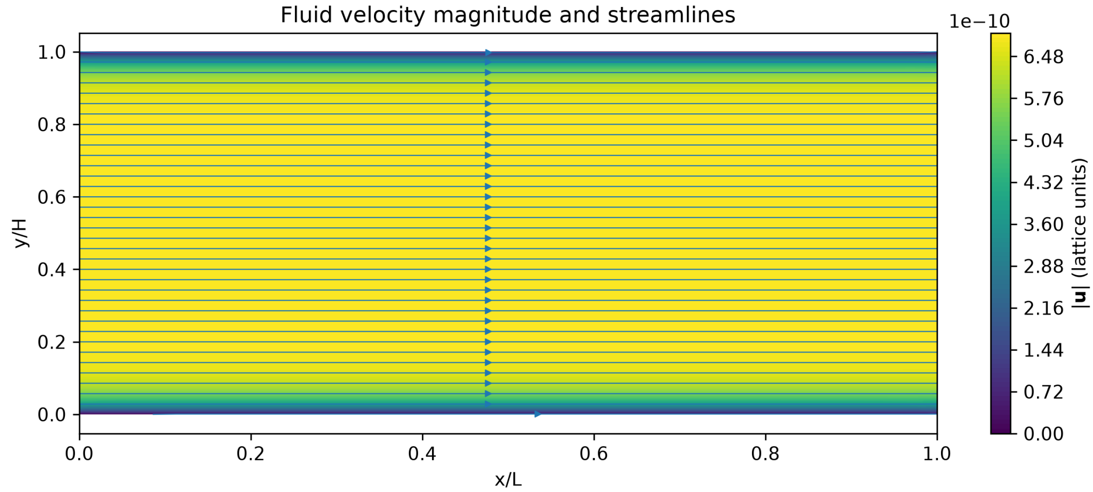


Figure 7: Electro-osmotic velocity field together with the corresponding streamlines.

# 8 Discussion

In the present work, a modular lattice Boltzmann framework has been developed for the simulation of coupled electrochemical transport phenomena. The framework combines three coupled lattice Boltz-

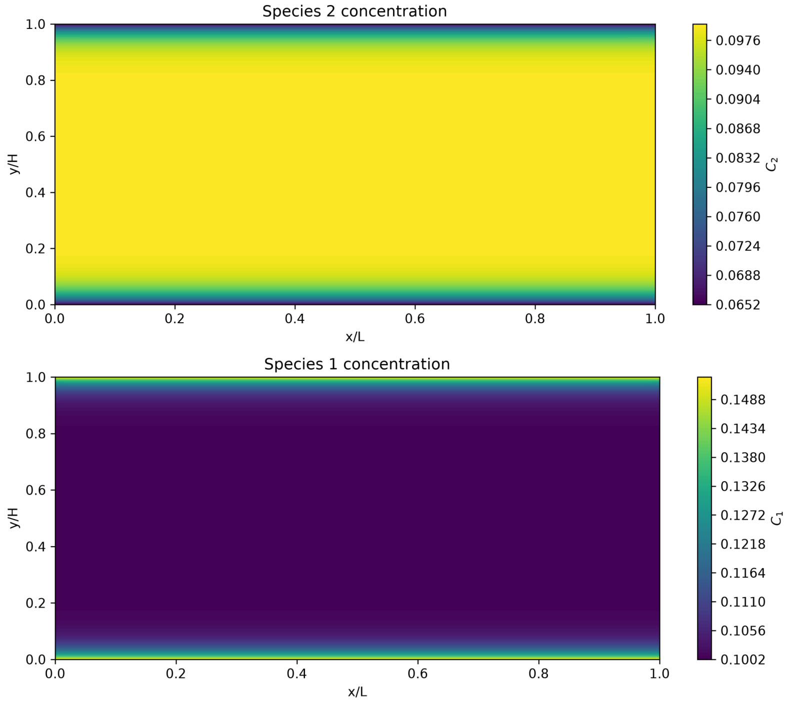


Figure 8: Contour distributions of the ionic concentrations within the computational domain.

mann solvers for the electrostatic potential, fluid flow, and ionic species transport within a unified numerical implementation. The primary objective of this contribution was to verify and validate each component of the numerical framework through analytical benchmark solutions. Consequently, a simple parallel-plate microchannel with well-defined physical and chemical boundary conditions was deliberately adopted to enable a rigorous assessment of the numerical accuracy of the proposed methodology.

Practical electrochemical systems are generally far more complex than the benchmark problem considered in this work. Real electrocatalytic reactors often involve complex geometries, porous catalyst layers, gas diffusion electrodes, heterogeneous and homogeneous chemical reactions, multicomponent electrolytes, and nonlinear boundary conditions. As a result, both the governing equations and the associated boundary conditions must be generalized to accurately describe realistic electrochemical devices. Nevertheless, establishing the numerical accuracy of the fundamental transport solvers under well-controlled conditions is an essential prerequisite before extending the framework to more complex systems.

Another important limitation of the present implementation is the description of ionic transport using the classical Nernst–Planck equation. While this formulation accurately describes dilute electrolyte solutions, it neglects steric and entropic effects arising from the finite size of ions[33–37]. Such effects become increasingly important near highly charged electrode surfaces, where ion crowding and excluded-volume interactions may significantly modify the electric double layer and the local ionic concentrations. Incorporating finite-size effects through modified transport equations or statistical thermodynamic models therefore represents an important direction for future developments.

The present contribution should therefore be viewed as the first stage in the development of a comprehensive lattice Boltzmann framework for reactive electrochemical transport. In future work, the geometrical restrictions will be removed by considering more realistic reactor configurations and complex boundary conditions. Furthermore, heterogeneous electrochemical reactions, homogeneous bulk reactions, and molecular-scale corrections associated with finite ion sizes will be incorporated into the numerical framework. These developments will ultimately enable predictive multiscale simulations of reactive electrochemical transport in realistic electrocatalytic systems.

## 9 Conclusion

In this work, a modular Fortran-based lattice Boltzmann framework was developed for the simulation of coupled electrochemical transport phenomena, including electrostatics, hydrodynamics, and ionic species transport. The framework employs three coupled lattice Boltzmann solvers to solve the Poisson, Navier–Stokes, and Nernst–Planck equations within a unified computational framework.

The proposed numerical methodology was systematically verified using analytical benchmark solutions for the electrostatic potential, electro-osmotic velocity, ionic concentration profiles, and the fully coupled electrokinetic system. Excellent agreement was obtained between the numerical and analytical solutions, demonstrating the accuracy and robustness of the implemented lattice Boltzmann formulation and the associated boundary conditions.

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## Conflicts of Interest

The authors declare no conflicts of interest.

## Data Availability Statement

The Fortran source code developed in this work is available from the corresponding author upon reasonable request. `https://github.com/ebrahim-tayyebi/lbm-electrokinetics`

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